

Using a Bayesian Network to Predict Term Deposit Subscription

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I. PROBLEM STATEMENT

Financial institutions face significant challenges in predicting customer behavior, specifically their tendency to subscribe to financial products like term deposits. Traditional methods of targeting customers are not precise and inefficient at reaching the specific customers who are most likely to sign up for things like term deposits [1]. This could cause wasted resources, frustrated customers, and missed opportunities. Also, the financial landscape is ever-changing, with economic conditions and customer preferences constantly evolving. Existing prediction models may struggle to adapt to these changes, leading to inaccurate results [2].

II. DOMAIN DEFINITION

This project is in the intersection of two important domains:

- **Financial Marketing:** This domain focuses on the strategies used by financial institutions to promote their products and services to potential and existing customers. Financial institutions are increasingly utilising data-driven approaches to gain insights into customer behaviour, improve decision-making, and enhance operational efficiency.
- **Customer Relationship Management:** This domain deals with the strategies used by businesses to improve customer interactions, personalise experiences, and promote loyalty. These domains connect to areas such as predictive modelling, statistical analysis, and probabilistic reasoning to improve business outcomes and strategic decision-making.

III. PURPOSE STATEMENT

The purpose of this assignment is to develop a predictive model using a Bayesian Network to forecast term deposit subscriptions for banking clients.

This project aims to:

- Analyse the relationships between different variables to inform us about factors that influence customer behaviour associated with term deposits. This can provide advantageous insights for banks to develop more effective marketing strategies and improve customer targeting.
- Show how Probabilistic Graphical Models can be effectively used for customer behavior prediction in the financial marketing domain. [2] and [3] show the use of Probabilistic Graphical Models for financial customer behavior prediction and highlight their potential for interpretable results.

IV. DATA HANDLING AND ETHICAL CONSIDERATIONS

A. Data Preprocessing

My code checks for missing values in the original dataset “bank.csv” [5], but it does not handle them as there are no missing values in the current dataset. This is observed when I print the number of missing values for each column and all columns output the value 0.

B. Removal of Unnecessary Features

The original dataset contained the following columns: ‘age’, ‘job’, ‘marital’, ‘education’, ‘default’, ‘balance’, ‘housing’, ‘loan’, ‘contact’, ‘day’, ‘month’, ‘duration’, ‘campaign’, ‘pdays’, ‘previous’, ‘poutcome’, ‘y’.

I do not consider the following features to avoid complexities and pointless relationships in the Bayesian network, thereby leading to reliable predictions:

- ‘age’: people of any age can decide whether to subscribe to a term deposit or not and they can all make different decisions.
- ‘contact’ (communication type): The way in which the bank contacted the customer previously is not directly relevant to future subscription decisions.
- ‘day of week’ (last contact day): The specific day of a past contact does not influence a customer’s future subscription decision.
- ‘month’ (last contact month): The month of a past contact has no influence on a customer’s future subscription decision.
- ‘duration’ (last contact duration): This information becomes available only after someone has been contacted, potentially creating an unrealistic prediction scenario. It does not influence a customer’s future subscription decision.

C. Converting Numerical Columns into Categorical Columns

I decided to change the numerical columns to categorical columns. Before doing this, I first scaled the numerical columns (so that each variable influences the target variable with the same weight). Then I discretised the values so that I would not have to calculate the probability for each value for the CPT and the numbers could be categorised into ranges. But then I figured that it would be easier to convert the numerical columns to categorical columns (a type of discretisation as I am putting continuous variables into ranges) as it would reduce the complexity of my code.

I defined functions for these columns that take in the

numerical values of the columns and return the categorical values, which are then placed into new columns. I split the ranges accordingly.

Benefits:

- Categorical features are easier to interpret and understand.
- Non-linear relationships between the features and the target variable are better captured.
- Categorical data may be less sensitive to outliers than numerical data- since outliers are grouped into their categories.

Limitations:

- I could have lost important information as the original numerical values are replaced with discrete categories- which could result in decreased model performance if numerical columns have crucial values.
- I could not have chosen the best way to categorise the numerical values- which could have caused changes in the decisions the model makes.

An alternative method would have been to ordinal encode the features, which could have possibly retained more information from the numerical values, but I preferred categorical features due to reduced complexity.

D. Renamed Columns

Then I renamed some columns to clarify what data the column contains they contain as their names in “bank.csv” are vague and can be misinterpreted.

E. Ethical Considerations

I have cited the repository I used for the dataset. I have ensured that the dataset I used does not contain any sensitive or personally identifiable financial data and no person’s privacy is harmed in any way.

F. Split data into training and testing set

The data is split into the training set and test set- 70:30. This allows us to assess the model’s performance on unseen data, which indicates how the model will generalize to new, real-world data. By training the model on the training set and evaluating its performance on the separate test set, we get an unbiased estimate of the model’s performance and avoid overfitting.

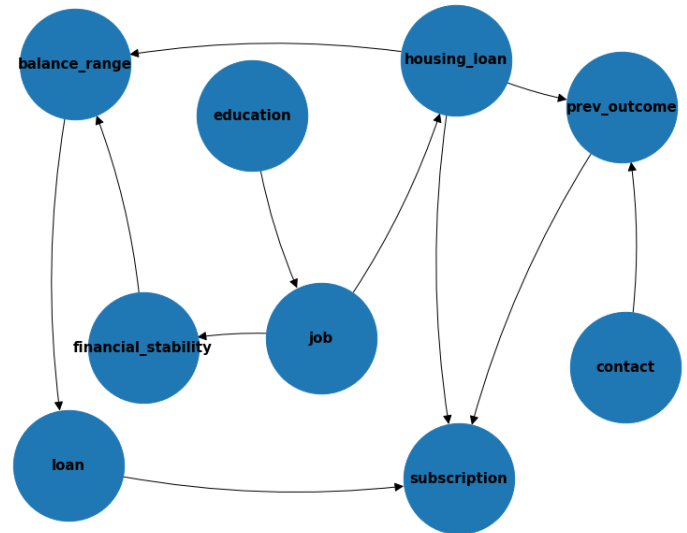
V. THE BAYESIAN NETWORK MODEL

A. Choice of Bayesian Network

Reasons I used a Bayesian Network:

- Provides a graphical representation of the probabilistic relationships between variables, making it easier to understand and interpret the model.
- Efficient at modelling the complex relationships between variables. Customer behaviour is influenced by various interconnected factors, and Bayesian Networks effectively capture these dependencies.
- Efficient at handling uncertainty, which is inherent in

Fig. 1. A graph representing the Bayesian Network



real-world data. This is valuable for predicting customer behaviour, which is often influenced by various factors beyond what the model can perfectly capture.

- Represents causal relationships between variables, allowing for causal reasoning and inference, which is beneficial in understanding the underlying mechanisms influencing the target variable ‘subscription’.
- Incorporates prior knowledge, which is extremely useful in the finance and marketing domains where data can be incomplete or noisy.
- The structure learning process in my code allows for incorporating domain knowledge as I manually remove unneeded nodes and edges- potentially improving the model’s interpretability and performance.
- I have used the Expectation-Maximization algorithm to estimate the latent variable ‘financial_stability’, showing the ability of Bayesian Networks to handle unobserved variables.

Limitations:

- Computationally expensive with several variables or complex relationships between variables. Learning the optimal structure for a complex Bayesian Network is time-consuming and computationally demanding. Thus, I lowered the number of variables in the model to simplify it.
- Hill-climbing (Structure Learning algorithm) may converge to a local optimum, and the learned structure may not be globally optimal, impacting the model’s performance.
- May not capture complex relationships or distributions effectively, especially in high-dimensional or highly non-linear problems. That is why I ensured that my problem is simple enough to be represented properly by a Bayesian Network.

Alternatives: 1. Markov Random Fields:

- Can model undirected dependencies between variables.
- Can handle cyclic dependencies between variables, which is

not possible in Bayesian Networks due to their acyclic nature.

- Effective for modeling spatial or structured data, like images or language data, where dependencies between neighboring variables are important.
- Do not make the strong conditional independence assumptions that Bayesian Networks do, which can be beneficial when the assumptions are violated.

However, I used Bayesian Networks over Markov Random Fields because:

- Two of my variables, 'job' and 'balance_range' had dependencies both ways, causing cyclic behavior, so I introduced a latent variable 'financial_stability' to get rid of the undirected dependency. Markov Random Fields would have been better to solve this issue but since there was only one dependency I decided to use a latent variable.
- My data does not have any spatial or properly structured data.
- Bayesian Networks can represent causal relationships between variables, which is not as straightforward in Markov Random Fields because of their undirected nature.
- The directed acyclic graph structure of Bayesian Networks provides better interpretability and understanding of the relationships between variables.

Chain Graphs (a generalisation of both Bayesian Networks and MRFs, combining directed and undirected edges):

- model a combination of both directed and undirected dependencies in a single graphical model.
- Can represent more complex dependency structures that cannot be easily captured by either Bayesian Networks or Markov Random Fields alone.
- Do not make the strong conditional independence assumptions of Bayesian Networks, which can be beneficial when the assumptions are violated.

However, I used Bayesian Networks over Chain Graphs because:

- My problem can be represented by a directed acyclic graph so Bayesian Networks are simpler and more interpretable than Chain Graphs.
- Bayesian Networks are more suitable for causal reasoning and inference since the relationships between variables in my model are mostly causal.
- Inference and learning algorithms for Bayesian Networks can be more efficient and scalable than those for Chain Graphs due to their complexity.

B. Evaluation Metrics

1) *Accuracy*: The accuracy of my model is 89.22%. This means that it correctly predicts the target variable ('subscription') in approximately 89.22% of the cases in the test set. This is a reasonably good accuracy score.

2) *Confusion matrix*: The rows represent the true classes, and the columns represent the predicted classes.

- The top-left value (11807) indicates the number of true negatives (correctly predicted as not subscribing).
- The top-right value (159) indicates the number of false positives (incorrectly predicted as subscribing when they did not).
- The bottom-left value (1303) indicates the number of false negatives (incorrectly predicted as not subscribing when they did).
- The bottom-right value (295) indicates the number of true positives (correctly predicted as subscribing).

From the confusion matrix, it is evident that my model performs better in predicting non-subscribers than subscribers.

3) *Classification Report*: • The precision for the non-subscriber class (0.90) is high, indicating a low false-positive rate.

- The recall for the non-subscriber class (0.99) is also high, indicating a low false-negative rate.
- However, the precision for the subscriber class (0.65) is relatively low, suggesting a higher false-positive rate for this class.
- The recall for the subscriber class (0.18) is quite low, indicating that the model struggles to identify true subscribers accurately.

The low precision and recall for the subscriber class could be due to class imbalance in the data, where the non-subscriber class dominates. ROC AUC The ROC AUC (Area Under the Receiver Operating Characteristic Curve) score of 0.586 is relatively low, indicating that my model's ability to distinguish between the two classes (subscribers and non-subscribers) is not significantly better than random guessing. A value of 0.5 represents a random classifier, thus a score of 0.586 is only slightly better than random.

4) *Curve Interpretations*: • Calibration Curve (Predicted Probability versus True Probability):

The calibration curve shows a relatively straight line, indicating that my model's predicted probabilities are matched with the true probabilities.

- ROC Curve (False Positive Rate versus True Positive Rate): The ROC curve shows a steep increase in the true positive rate initially, followed by a more gradual increase as the false positive rate increases. This indicates that my model can capture a portion of the positive class (subscribers) relatively well, but its performance deteriorates as the false positive rate increases.

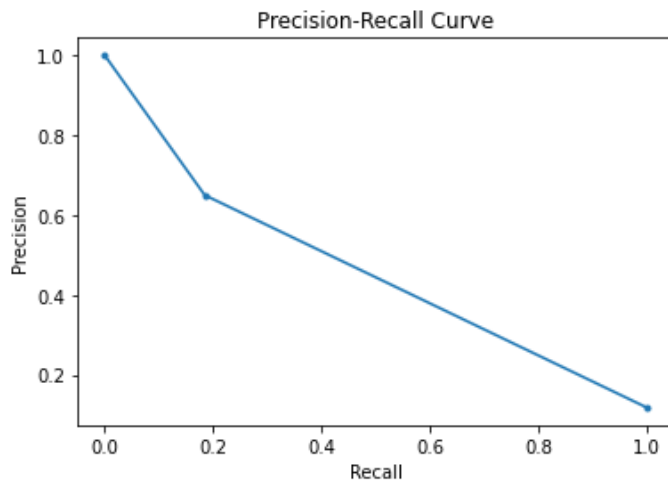
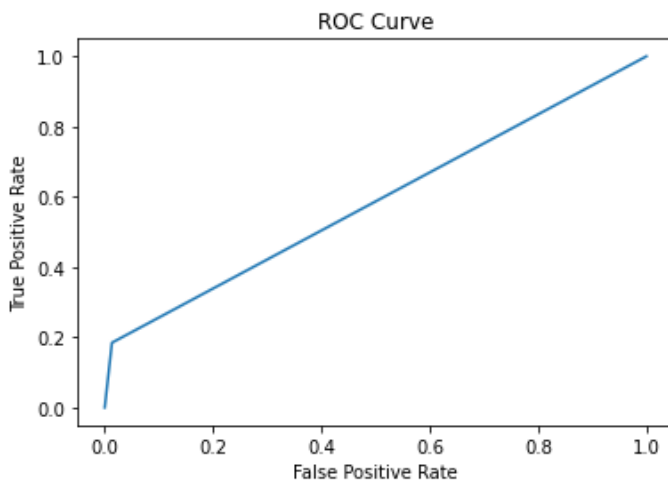
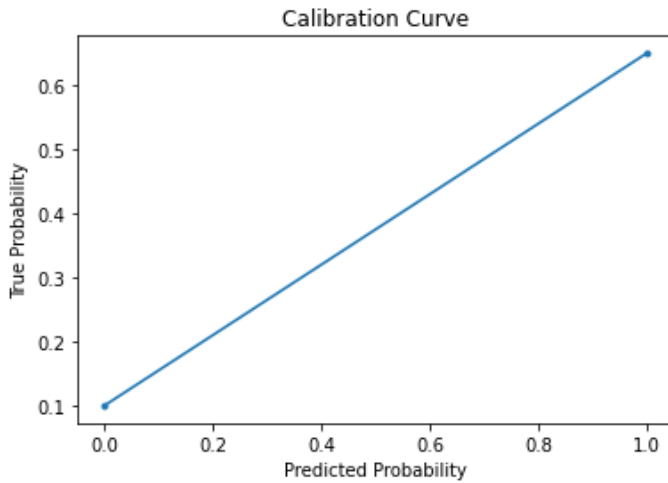
• Precision-Recall Curve (Recall versus Precision):

The precision-recall curve shows a sharp drop in precision as recall increases, particularly in the initial region (recall up to 0.2). This further confirms the model's difficulty in maintaining high precision while increasing recall for the positive class (subscribers).

VI. INFERENCE TECHNIQUES

I used exact inference because:

- In the financial and marketing domains, decisions can have significant economic and business implications. Accurate



predictions of customer behavior, in my model's case-subscription to term deposits, can directly impact revenue, customer retention, and overall profitability. Exact inference ensures that the probabilities and predictions are calculated accurately, minimizing the risk of errors that could result in

suboptimal decisions.

- In financial and marketing data, small changes in input variables or probabilities can have significant impacts on outcomes. Exact inference algorithms accurately capture these small changes. This provides precise updates to the probabilities and predictions. This is crucial for making informed decisions using the most up-to-date and accurate information.
- Financial and marketing data have intricate dependencies between various features. Exact inference algorithms can effectively handle these complex dependencies within the Bayesian Network structure, ensuring that all relevant information is accurately incorporated into the predictions.

I used Variable Elimination for:

1. Inferring the posterior probabilities of the latent variable 'financial_stability' given the observed data (evidence) during the Expectation-Maximization algorithm.

2. Performing inference on the test data to predict the subscription likelihood for each instance.

Benefits:

- Provides a precise and efficient way to compute the marginal probabilities of the latent variable given the observed evidence, which is important for the Expectation-Maximization algorithm's convergence and accuracy.
- Provides a precise and interpretable way to compute the conditional probabilities of the target variable ('subscription') given the evidence from the test data, leading to accurate predictions and decision-making.
- Results are transparent, allowing us to understand and explain the reasoning behind the predictions, which is crucial in domains like finance and marketing.

Limitations:

- As the number of variables and complexity of relationships increase in the PGM, Variable Elimination can become computationally expensive, especially for several features, and cause the slowing down of the Expectation-Maximization convergence. Fortunately, my problem does not have too many features.

VII. LEARNING TECHNIQUES

A. Structure Learning

I used the Hill-Climbing algorithm with the Bayesian Information Criterion (BIC) score to learn the structure of the Bayesian Network from the training data. The Hill-Climbing algorithm is efficient and quickly finds a good solution. However, if it gets stuck in local optima, it will not be able to find the globally optimal structure- this does not happen with my model so the algorithm is effective to use. By using the BIC score, I found a structure with a reasonable trade-off between model complexity and data fit, helping to avoid overfitting.

After learning the initial structure, I manually removed certain nodes and edges from the learned structure based on domain knowledge or assumptions. This simplified the model as it removed unnecessary dependencies, potentially improving interpretability and generalization capabilities. These are the nodes I removed after structure learning:

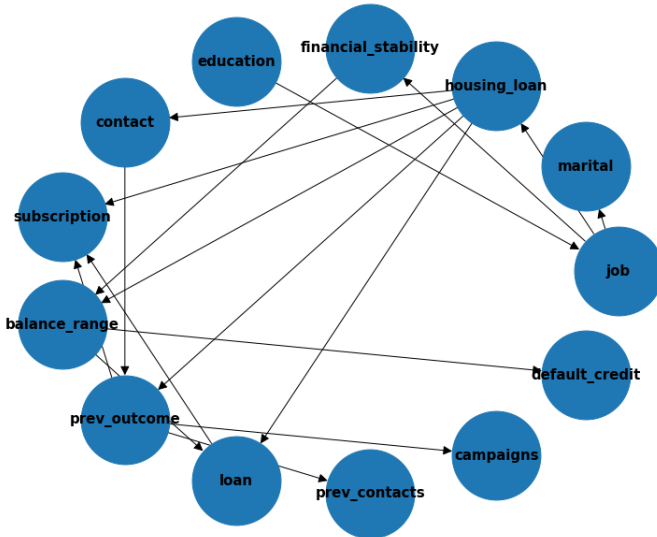
- 'marital' node because it only had arrows coming to it (from 'age_range' and 'education'- directly influenced by them) and no arrows going from it.
- 'default_credit' node because it had one arrow coming to it from 'balance_range' (directly influenced by it) but no arrows going from it.
- 'num_campaign_contacts' node because it had one arrow coming to it from 'prev_outcome' (directly influenced by it) but no arrows going from it.
- 'prev_campaign_contacts' node because it has one arrow coming to it from 'prev_outcome' (directly influenced by it) but no arrows going from it.

The above nodes do not influence any factors- most importantly they do not influence our target variable 'subscription' either directly or indirectly.

These are the edges I removed after structure learning:

- 'housing_loan' to 'past_contact': it is unclear how the housing loan status would directly influence the past contacts made to a person. There is no explainable dependency.
- 'housing_loan' to 'loan': having a housing loan does not necessarily correlate strongly with having a personal loan. While there might be some overlap between individuals who have both types of loans, there could also be individuals who have one type of loan but not the other.

Fig. 2. A graph representing the Bayesian Network before removing unnecessary nodes and edges



B. Parameter Learning

I used the Bayesian Dirichlet Equivalent Uniform (BDeu) prior to estimate the conditional probability distributions (CPDs) of the Bayesian Network.

Benefits:

- Helped to avoid overfitting, especially for dealing with small sample sizes or sparse data (in my case), by incorporating prior information.
- Produced smoother and more regularized parameter estimates, reducing the impact of noise or outliers in the data
- Allowed for incorporating domain knowledge (prior beliefs) through the choice of the equivalent sample size parameter.
- Provided a balanced trade-off between the prior information and the observed data.
- Its hyperparameters have clear meanings, facilitating understanding of the model's assumptions.

The choice of the equivalent sample size parameter impacted the strength of the prior. I experimented with different sample sizes (sensitivity analysis) and found that 10 is the optimal value (highest accuracy) for my problem.

Results:

Equivalent sample size: 1, Accuracy: 0.8922

Equivalent sample size: 5, Accuracy: 0.8925

Equivalent sample size: 10, Accuracy: 0.8927

Equivalent sample size: 20, Accuracy: 0.8923

Equivalent sample size: 50, Accuracy: 0.8920

Equivalent sample size: 100, Accuracy: 0.8918

The does not monotonically increase or decrease with the equivalent sample size, as the relationship between the prior strength and the model's performance can be complex and problem-specific. Therefore, I chose the one providing the best trade-off between the prior's influence and the model's fit to the data.

While the BDeu prior assumes a Dirichlet distribution for the parameters, this assumption may not accurately reflect the true distribution in all cases. Alternatively, I could have used the K2 prior. It is particularly useful when there is a lack of prior knowledge about the structure or parameters of the Bayesian Network. But since I learned the structure of the model, I used BDeu.

VIII. REFERENCES

REFERENCES

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- [4] Safarkhani, Fereshteh, and Sérgio Moro. "Improving the accuracy of predicting bank depositor's behavior using a decision tree." *Applied Sciences* 11, no. 19 (2021): 9016.
- [5] Moro,S., Rita,P., and Cortez,P.. (2012). Bank Marketing. UCI Machine Learning Repository. <https://doi.org/10.24432/C5K306>.